

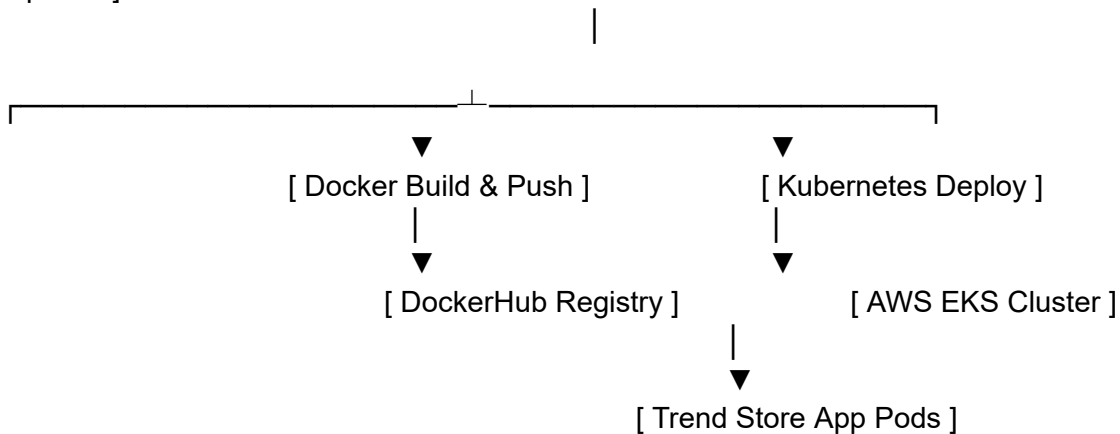
Final Project 2 - Trend Store Application

A production-ready React application (trend-store-app) packaged using Nginx, containerized with Docker, and deployed to an AWS EKS (Kubernetes) cluster via an automated Jenkins CI/CD pipeline. Complete with host and container monitoring via Prometheus and Grafana.

Project structure: <https://github.com/kousick-devaraj/Trend-store-app.git>

Architecture Overview

[Developer Push] —> [GitHub Repo] —(Webhook Trigger)—> [Jenkins CI/CD Pipeline]



Tech Stack

- **Frontend:** React (Pre-compiled dist static assets)
- **Web Server:** Nginx (alpine)
- **Containerization:** Docker
- **Infrastructure as Code (IaC):** Terraform
- **Cloud Provider:** AWS (VPC, EC2, EKS)
- **CI/CD Orchestration:** Jenkins
- **Container Orchestration:** AWS EKS (Kubernetes)
- **Monitoring:** Prometheus, Node Exporter, cAdvisor & Grafana

Clone repo and create your own repo to push all files

```
Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project2/Trend-Store-App (master)
$ git branch -M main

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project2/Trend-Store-App (main)
$ git remote add origin https://github.com/kousick-devaraj/Trend-store-app.git

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project2/Trend-Store-App (main)
$ git push -u origin main
Enumerating objects: 78, done.
Counting objects: 100% (78/78), done.
Delta compression using up to 8 threads
Compressing objects: 100% (78/78), done.
Writing objects: 100% (78/78), 8.58 MiB | 4.83 MiB/s, done.
Total 78 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), done.
To https://github.com/kousick-devaraj/Trend-store-app.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```

1. Local Containerization & Verification

The Docker container serves static files using Nginx listening on port 80 internally, while exposing host port 3000.

Build the Docker image

```
docker build -t kousickd/trend-store-app:v1 .
```

Run container locally on port 3000

```
docker run -d -p 3000:80 --name trend-app kousickd/trend-store-app:v1
```

```
MINGW64/c/Users/Kousick Devaraj/Desktop/GuviProjects/Project2/Trend-Store-App
Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project2/Trend-Store-App (main)
$ ls -lrt
total 5
-rw-r--r-- 1 Kousick Devaraj 197121 1801 Sep 13 12:19 README.md
drwxr-xr-x 1 Kousick Devaraj 197121 0 Sep 13 12:19 dist/
-rw-r--r-- 1 Kousick Devaraj 197121 250 Sep 13 12:42 Dockerfile

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project2/Trend-Store-App (main)
$ docker build -t trend-store-app:v1 .
[+] Building 21.1s (8/8) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 289B
=> [internal] load metadata for docker.io/library/nginx:alpine
=> [auth] library/nginx:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [internal] load build context
=> => transferring context: 9.24MB
=> CACHED [1/2] FROM docker.io/library/nginx:alpine@sha256:72ba65eb42c10344912a84ff42408db7d34f2feb642204570ab8fc5ffd29f1d3
=> => resolve docker.io/library/nginx:alpine@sha256:72ba65eb42c10344912a84ff42408db7d34f2feb642204570ab8fc5ffd29f1d3
=> [2/2] COPY dist /usr/share/nginx/html
=> exporting to image
=> exporting layers
=> => exporting manifest sha256:234b0676926d19ec0d5d29cfa9d9247d409f50d56f18428c110965d2d0d3393e
=> => exporting config sha256:cf575ebdf411e222abbadc9be357fdc3b8ac01be42d56a9614d104eda12d6cba
=> => exporting attestation manifest sha256:af70b8014ecde9a878c3a76b5556eaba57c2973a75d33bea3d3325954468799f
=> => exporting manifest list sha256:40c96cd7011535ed40d852c1b44d05eee6492c3dffbbdb6fcd8a59068a2e9421e
=> naming to docker.io/library/trend-store-app:v1
=> unpacking to docker.io/library/trend-store-app:v1

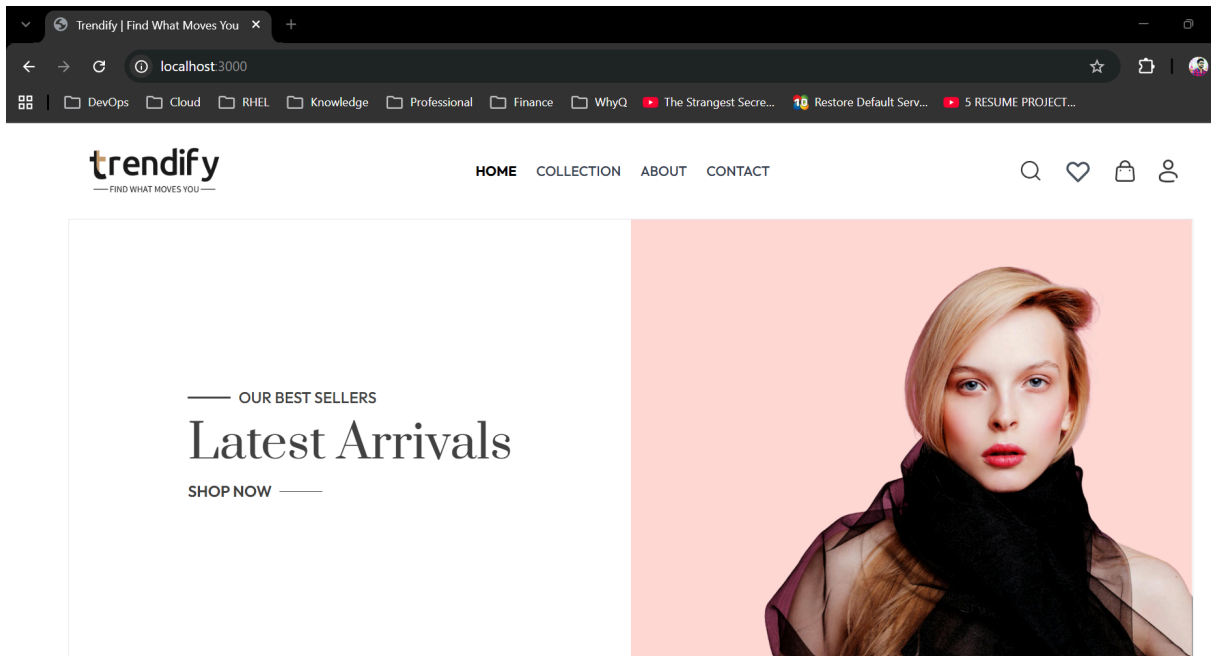
View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/7giri9ym06pv0gqgg0if42vi

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project2/Trend-Store-App (main)
$ docker run -d -p 3000:80 --name trend-app trend-store-app:v1
eb1cea29b28683171b3c5171736e0dc8b3875fb83e0aad7bd896c10a7c3b9c21

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project2/Trend-Store-App (main)
$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
eb1cea29b286   trend-store-app:v1  "/docker-entrypoint..."  4 seconds ago  Up 4 seconds  0.0.0.0:3000->80/tcp, [::]:3000->80/tcp  trend-app

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project2/Trend-Store-App (main)
$ |
```

Application running fine on local. Verify application locally at <http://localhost:3000>.



2. Infrastructure Provisioning (Terraform)

Provision the underlying AWS VPC and Jenkins EC2 server:

```
# Initialize and apply Terraform config
```

```
terraform init
```

```
terraform apply --auto-approve
```

Provision the AWS EKS Cluster via eksctl:

Bash

```
eksctl create cluster \
```

```
--name trend-eks-cluster \
```

```
--region us-east-1 \
```

```
--nodegroup-name standard-workers \
```

```
--node-type t3.medium \
```

```
--nodes 2
```

```
MINGW64/c/Users/Kousick Devaraj/Desktop/GuviProjects/Project2/Trend-Store-App/terraform
-rw-r--r-- 1 Kousick Devaraj 197121 250 Sep 13 12:42 Dockerfile
drwxr-xr-x 1 Kousick Devaraj 197121 0 Sep 13 14:53 terraform/
drwxr-xr-x 1 Kousick Devaraj 197121 0 Sep 13 14:53 ./

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project2/Trend-Store-App (main)
$ cd terraform/

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project2/Trend-Store-App/terraform (main)
$ ls -lrt
total 0

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project2/Trend-Store-App/terraform (main)
$ vi main.tf

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project2/Trend-Store-App/terraform (main)
$ ls -lrt
total 4
-rw-r--r-- 1 Kousick Devaraj 197121 1833 Sep 13 14:57 main.tf

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project2/Trend-Store-App/terraform (main)
$ pdw
bash: pdw: command not found

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project2/Trend-Store-App/terraform (main)
$ terraform init
Initializing provider plugins found in the configuration...
- Finding latest version of hashicorp/aws...
- Installing hashicorp/aws v6.64.0...
- Installed hashicorp/aws v6.64.0 (signed by HashiCorp)

Initializing the backend...

Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
```

```
MINGW64/c/Users/Kousick Devaraj/Desktop/GuviProjects/Project2/Trend-Store-App/terraform
+ map_public_ip_on_launch = true
+ owner_id                 = (known after apply)
+ private_dns_hostname_type_on_launch = (known after apply)
+ region                   = "us-east-1"
+ tags                     = {
+   + "Name" = "trend-public-subnet"
+ }
+ tags_all                 = {
+   + "Name" = "trend-public-subnet"
+ }
+ vpc_id                   = (known after apply)
}

# aws_vpc.main_vpc will be created
+ resource "aws_vpc" "main_vpc" {
+   arn = (known after apply)
+   cidr_block = "10.0.0.0/16"
+   default_network_acl_id = (known after apply)
+   default_route_table_id = (known after apply)
+   default_security_group_id = (known after apply)
+   dhcp_options_id = (known after apply)
+   enable_dns_hostnames = true
+   enable_dns_support = true
+   enable_network_address_usage_metrics = (known after apply)
+   id = (known after apply)
+   instance_tenancy = "default"
+   ipv6_association_id = (known after apply)
+   ipv6_cidr_block = (known after apply)
+   ipv6_cidr_block_network_border_group = (known after apply)
+   main_route_table_id = (known after apply)
+   owner_id = (known after apply)
+   region = "us-east-1"
+   tags = {
+     + "Name" = "trend-vpc"
+   }
+   tags_all = {
+     + "Name" = "trend-vpc"
+   }
+ }

Plan: 7 to add, 0 to change, 0 to destroy.

Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to take exactly these actions if you run "terraform apply" now.

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project2/Trend-Store-App/terraform (main)
$ |
```

```
MINGW64/c/Users/Kousick Devaraj/Desktop/GuviProjects/Project2/Trend-Store-App/terraform
tags_all = {
  "Name" = "trend-public-subnet"
}
vpc_id = "vpc-02644bef43af48674"
}

# aws_vpc.main_vpc:
resource "aws_vpc" "main_vpc" {
  arn = "arn:aws:ec2:us-east-1:786352305260:vpc/vpc-02644bef43af48674"
  assign_generated_ipv6_cidr_block = false
  cidr_block = "10.0.0.0/16"
  default_network_acl_id = "acl-083f8bb8a80972bdd"
  default_route_table_id = "rtb-0ac810aace8b5f6d7"
  default_security_group_id = "sg-09c56bb863684b5ae"
  dhcp_options_id = "dopt-0091fe6d92162302f"
  enable_dns_hostnames = true
  enable_dns_support = true
  enable_network_address_usage_metrics = false
  id = "vpc-02644bef43af48674"
  instance_tenancy = "default"
  ipv6_association_id = null
  ipv6_cidr_block = null
  ipv6_cidr_block_network_border_group = null
  ipv6_ipam_pool_id = null
  ipv6_netmask_length = 0
  main_route_table_id = "rtb-0ac810aace8b5f6d7"
  owner_id = "786352305260"
  region = "us-east-1"
  tags = {
    "Name" = "trend-vpc"
  }
  tags_all = {
    "Name" = "trend-vpc"
  }
}

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project2/Trend-Store-App/terraform (main)
$ terraform state list
aws_instance.jenkins_server
aws_internet_gateway.igw
aws_route_table.rt
aws_route_table_association.rta
aws_security_group.jenkins_sg
aws_subnet.public_subnet
aws_vpc.main_vpc

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project2/Trend-Store-App/terraform (main)
$ |
```

Validate resources from AWS console:

Instances (1/1) Info Info Last updated 4 minutes ago Connect Instance state Actions Launch instances

Saved filter sets Choose filter set

Instance state: running Clear filters

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs
Jenkins-Server	i-0dca6995f673f960b	Running	t3.medium	3/3 checks passed	us-east-1a	ec2-3-235-157-253.co...	3.235.157.253	-	-

EC2 EC2 > Instances > i-0dca6995f673f960b

Dashboard
 AWS Global View
 Events
 Instances
 Instances
 Instance Types
 Launch Templates
 Spot Requests
 Savings Plans
 Reserved Instances
 Dedicated Hosts
 Capacity Reservations
 Capacity Manager
 Images
 AMIs
 AMI Catalog
 Elastic Block Store
 Volumes
 Snapshots
 Lifecycle Manager
 Network & Security
 Security Groups
 Elastic IPs

Details | **Status and alarms** | **Monitoring** | **Security** | **Networking** | **Storage** | **Tags**

▼ Security details

IAM role
 -
 Owner ID
 786352305260
 Launch time
 Sun Sep 13 2026 15:14:02 GMT+0530 (India Standard Time)

Security groups
 sg-08902378ab3a0fdca (jenkins-sg)

▼ Inbound rules

Name	Security group rule ID	Port range	Protocol	Source	Security groups	Description
-	sg-0c10c341267be30f5	8080	TCP	0.0.0.0/0	jenkins-sg	-
-	sg-0b104c2d0dd603560	22	TCP	0.0.0.0/0	jenkins-sg	-
-	sg-017b14121f0a1a52f	3000	TCP	0.0.0.0/0	jenkins-sg	-

▼ Outbound rules

Name	Security group rule ID	Port range	Protocol	Destination	Security groups	Description
-	sg-028bf509b0148ee2f	All	All	0.0.0.0/0	jenkins-sg	-

VPC VPC > Your VPCs

PC dashboard
 Filter by VPC
 Virtual private cloud
 Your VPCs
 Subnets
 Route tables
 Internet gateways
 Egress-only internet gateways

Your VPCs (2) Info Info Last updated 4 minutes ago Actions Create VPC

Name	VPC ID	State	Encryption c...	Encryption control ...	Block Public...	IPv4 CIDR	IPv6 CIDR
trend-vpc	vpc-02644bdf43af48674	Available	-	-	Off	10.0.0.0/16	-
-	vpc-0b9cdf14a562b637	Available	-	-	Off	172.31.0.0/16	-

VPC VPC > Subnets

VPC dashboard
 Filter by VPC
 Virtual private cloud
 Your VPCs
 Subnets
 Route tables
 Internet gateways
 Egress-only internet gateways

Subnets (7) Info Info Last updated 4 minutes ago Actions Create subnet

Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR	IPv6 CIDR
trend-public-subnet	subnet-01d3f6e1ab400672a	Available	vpc-02644bdf43af48674 trend-vpc	Off	10.0.1.0/24	-
-	subnet-01242640c00988191	Available	vpc-0b9cdf14a562b637	Off	172.31.64.0/20	-
-	subnet-04b7621924dcd853d	Available	vpc-0b9cdf14a562b637	Off	172.31.32.0/20	-
-	subnet-010f31c053406e315	Available	vpc-0b9cdf14a562b637	Off	172.31.80.0/20	-
-	subnet-002556a8bef2d99ac	Available	vpc-0b9cdf14a562b637	Off	172.31.48.0/20	-
-	subnet-0908f059e35991c04	Available	vpc-0b9cdf14a562b637	Off	172.31.16.0/20	-
-	subnet-0cea98cd2715bde0d	Available	vpc-0b9cdf14a562b637	Off	172.31.0.0/20	-

VPC VPC > Internet gateways

VPC dashboard
 Filter by VPC
 Virtual private cloud
 Your VPCs
 Subnets
 Route tables
 Internet gateways
 Egress-only internet gateways

Internet gateways (2) Info Info Last updated 3 minutes ago Actions Create internet gateway

Name	Internet gateway ID	State	VPC ID	Owner
trend-igw	igw-03164072b31be9a45	Attached	vpc-02644bdf43af48674 trend-vpc	786352305260
-	igw-0b9cdf14a562b637	Attached	vpc-0b9cdf14a562b637	786352305260

VPC VPC > Route tables

VPC dashboard
 Filter by VPC
 Virtual private cloud
 Your VPCs
 Subnets
 Route tables
 Internet gateways
 Egress-only internet gateways

Route tables (3) Info Info Last updated 3 minutes ago Actions Create route table

Name	Route table ID	Explicit subnet associations	Edge associations	Main	VPC	Owner ID
-	rtb-0ac810aace8b5f6d7	-	-	Yes	vpc-02644bdf43af48674 trend-vpc	786352305260
-	rtb-0d2773d046a5235900	-	-	Yes	vpc-0b9cdf14a562b637	786352305260
-	rtb-075a5d9ac4f3d6562	subnet-01d3f6e1ab400672a / trend-public-subnet	-	No	vpc-02644bdf43af48674 trend-vpc	786352305260

3. CI/CD Pipeline Configuration (Jenkins)

Prerequisites & Setup:

1. **Plugins Installed:** Git, GitHub Integration, Docker Pipeline, Kubernetes CLI.
2. **Credentials Configured:** dockerhub-credentials added to Jenkins Global Credentials.
3. **EKS Access:** jenkins user authenticated to EKS using aws eks update-kubeconfig.
4. **GitHub Webhook:** Configured Payload URL
http://<JENKINS_EC2_IP>:8080/github-webhook/.

Pipeline Stages (Jenkinsfile):

- **Checkout Code:** Pulls latest code on commit trigger.
- **Build Docker Image:** Containerizes static files tagged with \${BUILD_NUMBER}.
- **Push to DockerHub:** Pushes tagged image and latest tag to DockerHub.
- **Deploy to AWS EKS:** Updates k8s/deployment.yaml dynamically and applies resources via kubectl.

Install required packages on the ec2 instance

```
aws
[Alt+S] Ask Amazon Q
United States (N. Virginia) kousickd (7)
ubuntu@ip-10-0-1-39:~$ history
1 clear
2 sudo apt update && sudo apt install -y openjdk-17-jdk docker.io unzip
3 curl https://awscli.amazonaws.com/awscli-exe-linux-x86_64.zip -o "awscli2.zip" && unzip awscli2.zip && sudo ./aws/install
4 curl -LO "https://dl.k8s.io/release/$curl -L -s https://dl.k8s.io/release/stable.txt)/bin/linux/amd64/kubectl" && chmod +x kubectl && sudo mv kubectl /usr/local/bin/
5 ls -lrt /usr/local/bin/kubectl
6 curl --silent --location "https://github.com/weaveworks/eksctl/releases/latest/download/eksctl_$(uname -s)_amd64.tar.gz" | tar xz -C /tmp && sudo mv /tmp/eksctl /usr/local/bin
7 ls -lrt /usr/local/bin/eksctl
8 aws --version
9 docker --version
10 kubectl --version
11 kubectl version
12 eksctl version
13 sudo apt update
14 sudo apt install fontconfig openjdk-21-jre
15 java -version
16 sudo wget -O /etc/apt/keyrings/jenkins-keyring.asc https://pkg.jenkins.io/debian-stable/jenkins.io-2026.key
17 echo "deb [signed-by=/etc/apt/keyrings/jenkins-keyring.asc] https://pkg.jenkins.io/debian-stable binary/" | sudo tee /etc/apt/sources.list.d/jenkins.list > /dev/null
18 sudo apt update
19 sudo apt install jenkins
20 clear
21 history
ubuntu@ip-10-0-1-39:~$
```

```
aws
[Alt+S] Ask Amazon Q
United States (N. Virginia) kousickd
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/MHEEL
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/METADATA
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/RECORD
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/top_level.txt
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/INSTALLER
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/licenses/AUTHORS.rst
inflating: aws/dist/prompt_toolkit-3.0.51.dist-info/licenses/LICENSE
You can now run: /usr/local/bin/aws --version
ubuntu@ip-10-0-1-39:~$ curl -LO "https://dl.k8s.io/release/$curl -L -s https://dl.k8s.io/release/stable.txt)/bin/linux/amd64/kubectl" && chmod +x kubectl && sudo mv kubectl /usr/local/bin/
% Total % Received % Xferd Average Speed Time Time Current
100 59.0M 100 59.0M 0 0 152M 0 --:--:-- --:--:-- --:--:-- 192M
ubuntu@ip-10-0-1-39:~$ ls -lrt /usr/local/bin/kubectl
-rwxrwxr-x 1 ubuntu ubuntu 6188626 Sep 13 12:04 /usr/local/bin/kubectl
ubuntu@ip-10-0-1-39:~$ curl --silent --location "https://github.com/weaveworks/eksctl/releases/latest/download/eksctl_$(uname -s)_amd64.tar.gz" | tar xz -C /tmp && sudo mv /tmp/eksctl /usr/local/bin
-rwxr-xr-x 1 ubuntu ubuntu 152608930 Aug 14 18:42 /usr/local/bin/eksctl
ubuntu@ip-10-0-1-39:~$ aws --version
aws-cli/2.36.44 Python/3.14.6 Linux/6.2.0-1017-aws exe/x86_64.ubuntu.22
ubuntu@ip-10-0-1-39:~$ docker --version
Docker version 29.1.3, build 29.1.3-0ubuntu3-22.04.2
ubuntu@ip-10-0-1-39:~$ kubectl --version
error: unknown flag: --version
See 'kubectl --help' for usage.
ubuntu@ip-10-0-1-39:~$ kubectl version
Client Version: v1.37.0
Kubernetes Version: v5.8.1
The connection to the server localhost:8080 was refused - did you specify the right host or port?
ubuntu@ip-10-0-1-39:~$ eksctl version
0.230.0
ubuntu@ip-10-0-1-39:~$
ubuntu@ip-10-0-1-39:~$
ubuntu@ip-10-0-1-39:~$
```

```
aws
[Alt+S] Ask Amazon Q
United States (N. Virginia)
ubuntu@ip-10-0-1-39:~$ java -version
openjdk version "21.0.12" 2026-07-21
OpenJDK Runtime Environment (build 21.0.12+8-1-22.04-Ubuntu)
OpenJDK 64-Bit Server VM (build 21.0.12+8-1-22.04-Ubuntu, mixed mode, sharing)
ubuntu@ip-10-0-1-39:~$ jenkins --version
2.568.3
ubuntu@ip-10-0-1-39:~$
```

Setting up jenkins configurations

← → ↺

⚠ Not secure 3.235.157.253:8080

☆

🔖

👤

🔍 | 📁 DevOps 📁 Cloud 📁 RHEL 📁 Knowledge 📁 Professional 📁 Finance 📁 WhyQ 📺 The Strangest Secre... 🧑 10 Restore Default Serv... 📺 5 RESUME PROJECT...

 **Jenkins**

+ New Item

📁 Build History

Build Queue

No builds in the queue.

Build Executor Status

0/2

🔍 ⚙️ 👤

Add description

Welcome to Jenkins!

This page is where your Jenkins jobs will be displayed. To get started, you can set up distributed builds or start building a software project.

Start building your software project

Create a job

Set up a distributed build

Set up an agent

Configure a cloud

Learn more about distributed builds

← → ↺

⚠ Not secure 3.235.157.253:8080

🔍 ☆

🔖

🔍 | 📁 DevOps 📁 Cloud 📁 RHEL 📁 Knowledge 📁 Professional 📁 Finance 📁 WhyQ 📺 The Strangest Secre... 🧑 10 Restore Default Serv... 📺 5 RESUME PROJECT...

 **Jenkins**

+ New Item

📁 Build History

Build Queue

No builds in the queue.

Build Executor Status

(0 of 2 executors busy)

All +

S	W	Name ↓	Last Success	Last Failure	Last Duration
🟢	☁	trend-store-app-pipeline	1 hr 7 min #3	3 hr 47 min #1	10 sec

Icons: S M L

←

→

↺

⚠ Not secure

3.235.157.253:8080/job/trend-store-app-pipeline/

🗂

DevOps

Cloud

RHEL

Knowledge

Professional

Finance

WhyQ

📺 The Stra

 **Jenkins** / trend-store-app-pipeline

Status

Changes

Build Now

Delete Pipeline

Configure

Rename

GitHub

Stages

Pipeline Syntax

GitHub Hook Log

🟢 trend-store-app-pipeline

Job to build, push and deploy react app using container in k8s on AWS.

Permalinks

- Last build (#3), 1 hr 8 min ago
- Last stable build (#3), 1 hr 8 min ago
- Last successful build (#3), 1 hr 8 min ago
- Last failed build (#1), 3 hr 48 min ago
- Last unsuccessful build (#1), 3 hr 48 min ago
- Last completed build (#3), 1 hr 8 min ago

Builds >

Filter

Today

🟢 #3 6:41 PM

🟢 #2 4:07 PM

🔴 #1 4:01 PM

←

→

↺

⚠ Not secure

3.235.157.253:8080/job/trend-store-app-pipeline/3/console

🗂

DevOps

Cloud

RHEL

Knowledge

Professional

Finance

WhyQ

📺 The Strangest Secre...

👤 Restore Default Serv...

📺 5 R

 **Jenkins** / trend-store-app-pipeline / #3

```
+ kubectl apply -f k8s/deployment.yaml
deployment.apps/trend-app-deployment created
[Pipeline] sh
+ kubectl apply -f k8s/service.yaml
service/trend-app-service unchanged
[Pipeline] }
[Pipeline] // script
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
Pipeline executed successfully and application deployed to EKS!
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

4. Kubernetes Service Access

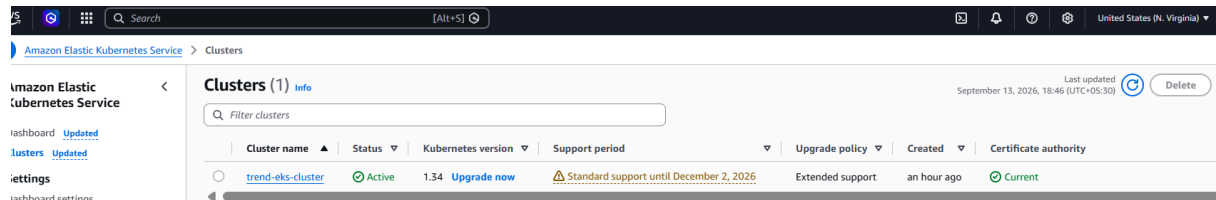
Deployments and Services route external traffic on port 3000 to the Nginx container port 80.

Check pod status

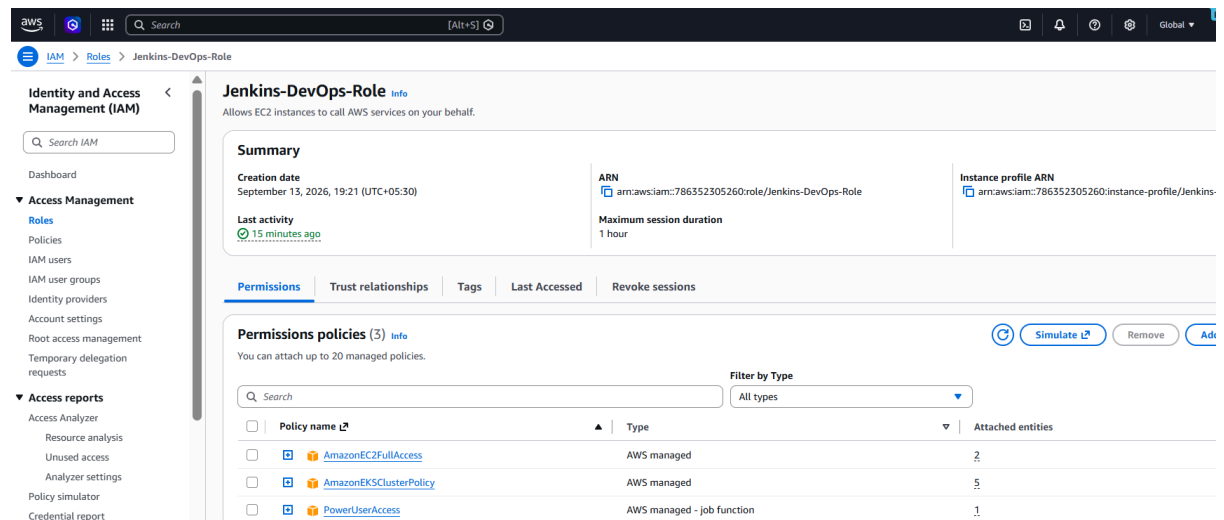
```
kubectl get pods -l app=trend-app
```

Get External IP of LoadBalancer

```
kubectl get svc trend-app-service
```

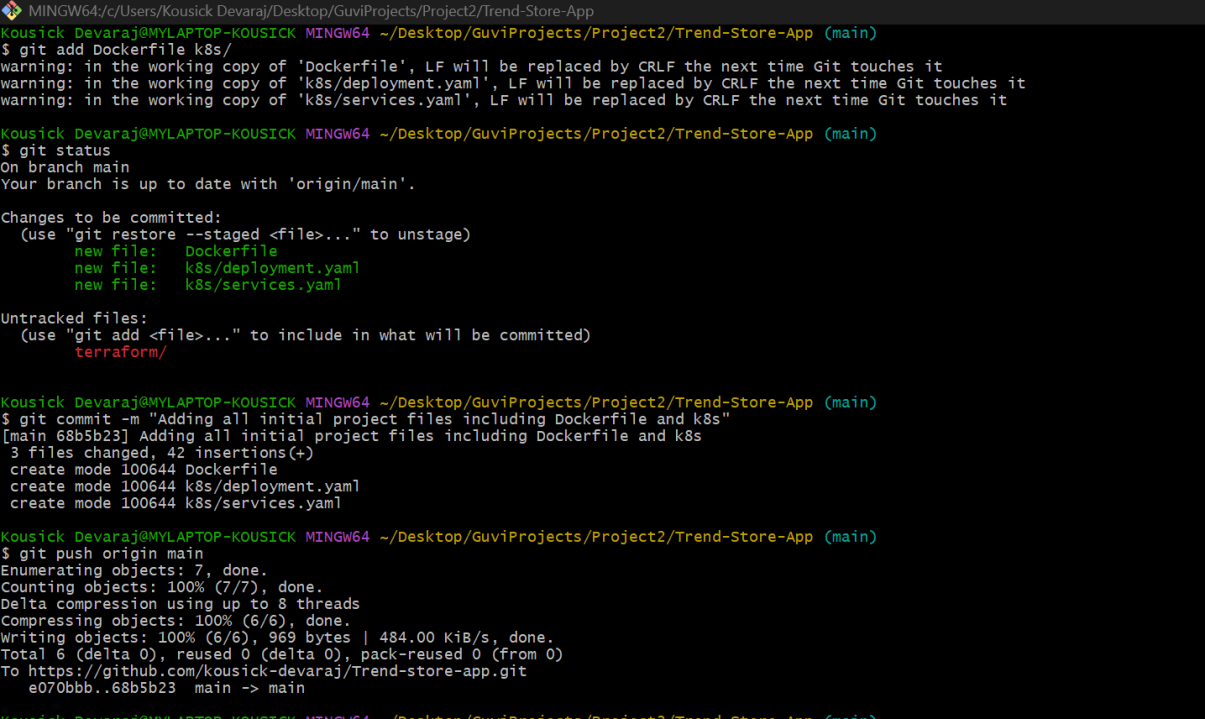


Role assigned to EC2 instance



Adding Dockerfile and k8s files to github

```
# git add .
# git status
# git restore --staged .
# git status
# git add Dockerfile k8s/
# git status
# git commit -m "Adding all initial project files including Dockerfile and k8s"
# git push origin main
```

A terminal window screenshot showing the process of adding Dockerfile and k8s files to a git repository. The user is in a MINGW64 environment at the path ~/Desktop/GuviProjects/Project2/Trend-Store-App. They run 'git add Dockerfile k8s/' which shows warnings about LF being replaced by CRLF. Then they run 'git status' showing the files are staged. Next, they run 'git commit -m "Adding all initial project files including Dockerfile and k8s"' which shows the commit details: 3 files changed, 42 insertions(+). Finally, they run 'git push origin main' which shows the push progress and the new commit hash e070bbb.68b5b23.

```
MINGW64/c/Users/Kousick Devaraj/Desktop/GuviProjects/Project2/Trend-Store-App
Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project2/Trend-Store-App (main)
$ git add Dockerfile k8s/
warning: in the working copy of 'Dockerfile', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'k8s/deployment.yaml', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'k8s/services.yaml', LF will be replaced by CRLF the next time Git touches it

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project2/Trend-Store-App (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   Dockerfile
        new file:   k8s/deployment.yaml
        new file:   k8s/services.yaml

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        terraform/

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project2/Trend-Store-App (main)
$ git commit -m "Adding all initial project files including Dockerfile and k8s"
[main 68b5b23] Adding all initial project files including Dockerfile and k8s
3 files changed, 42 insertions(+)
create mode 100644 Dockerfile
create mode 100644 k8s/deployment.yaml
create mode 100644 k8s/services.yaml

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project2/Trend-Store-App (main)
$ git push origin main
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 8 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 969 bytes | 484.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/kousick-devaraj/Trend-store-app.git
   e070bbb..68b5b23  main -> main
```

Adding terraform files:

```
# git status
# git add monitoring/
# git status
# git commit -m "Adding monitoring setup configuration files"
# git push origin main
```

```

MINGW64/c/Users/Kousick Devaraj/Desktop/GuviProjects/Project2/Trend-Store-App
Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project2/Trend-Store-App (main)
$ ls -lart terraform/
total 48
drwxr-xr-x 1 Kousick Devaraj 197121 0 Sep 13 14:58 .terraform/
-rw-r--r-- 1 Kousick Devaraj 197121 1452 Sep 13 14:58 .terraform.lock.hcl
-rw-r--r-- 1 Kousick Devaraj 197121 1836 Sep 13 15:12 main.tf
-rw-r--r-- 1 Kousick Devaraj 197121 11241 Sep 13 15:14 terraform.tfstate.backup
-rw-r--r-- 1 Kousick Devaraj 197121 16584 Sep 13 15:14 terraform.tfstate
drwxr-xr-x 1 Kousick Devaraj 197121 0 Sep 13 15:14 ./
drwxr-xr-x 1 Kousick Devaraj 197121 0 Sep 13 18:02 ../

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project2/Trend-Store-App (main)
$
Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project2/Trend-Store-App (main)
$ cat <<EOF > .gitignore
# Ignore local Terraform runtime and dependency files
terraform/.terraform/
terraform/.terraform.lock.hcl
terraform/*.tfstate
terraform/*.tfstate.backup
EOF

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project2/Trend-Store-App (main)
$ ls -lrt
total 9
-rw-r--r-- 1 Kousick Devaraj 197121 1801 Sep 13 12:19 README.md
drwxr-xr-x 1 Kousick Devaraj 197121 0 Sep 13 12:19 dist/
-rw-r--r-- 1 Kousick Devaraj 197121 250 Sep 13 12:42 Dockerfile
drwxr-xr-x 1 Kousick Devaraj 197121 0 Sep 13 15:14 terraform/
drwxr-xr-x 1 Kousick Devaraj 197121 0 Sep 13 18:06 k8s/

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project2/Trend-Store-App (main)
$ ls -lart
total 18
drwxr-xr-x 1 Kousick Devaraj 197121 0 Sep 13 12:16 ../
-rw-r--r-- 1 Kousick Devaraj 197121 1801 Sep 13 12:19 README.md
drwxr-xr-x 1 Kousick Devaraj 197121 0 Sep 13 12:19 dist/
-rw-r--r-- 1 Kousick Devaraj 197121 250 Sep 13 12:42 Dockerfile
drwxr-xr-x 1 Kousick Devaraj 197121 0 Sep 13 15:14 terraform/
drwxr-xr-x 1 Kousick Devaraj 197121 0 Sep 13 18:06 k8s/
drwxr-xr-x 1 Kousick Devaraj 197121 0 Sep 13 21:02 .git/
drwxr-xr-x 1 Kousick Devaraj 197121 0 Sep 13 21:07 ./
-rw-r--r-- 1 Kousick Devaraj 197121 153 Sep 13 21:07 .gitignore

```

```

MINGW64/c/Users/Kousick Devaraj/Desktop/GuviProjects/Project2/Trend-Store-App
Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project2/Trend-Store-App (main)
$ cat .gitignore
# Ignore local Terraform runtime and dependency files
terraform/.terraform/
terraform/.terraform.lock.hcl
terraform/*.tfstate
terraform/*.tfstate.backup

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project2/Trend-Store-App (main)
$ git add terraform/main.tf
warning: in the working copy of 'terraform/main.tf', LF will be replaced by CRLF the next time Git touches it

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project2/Trend-Store-App (main)
$ git add .gitignore
warning: in the working copy of '.gitignore', LF will be replaced by CRLF the next time Git touches it

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project2/Trend-Store-App (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   .gitignore
        new file:   terraform/main.tf

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project2/Trend-Store-App (main)
$ git commit -m "Add terraform folder with main.tf and configure terraform gitignore"
[main 4c777d1] Add terraform folder with main.tf and configure terraform gitignore
 2 files changed, 81 insertions(+)
 create mode 100644 .gitignore
 create mode 100644 terraform/main.tf

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project2/Trend-Store-App (main)
$ git push origin main
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (5/5), 1.07 KiB | 1.07 MiB/s, done.
Total 5 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/kousick-devaraj/Trend-store-app.git
 68b5b23..4c777d1  main -> main

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project2/Trend-Store-App (main)

```

```

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project2/Trend-Store-App (main)
$ git commit -m "Updated the dockerhub repo name mismatch"
[main 1d45b01] Updated the dockerhub repo name mismatch
1 file changed, 1 insertion(+), 1 deletion(-)

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project2/Trend-Store-App (main)
$ git push origin main
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 408 bytes | 408.00 KiB/s, done.
Total 4 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/kousick-devaraj/Trend-store-app.git
   c2b2af2..1d45b01  main -> main

Kousick Devaraj@MYLAPTOP-KOUSICK MINGW64 ~/Desktop/GuviProjects/Project2/Trend-Store-App (main)
$

```

Jenkins console URL: <http://3.235.157.253:8080>

Webhook URL: <https://3.235.157.253:8080/github-webhook>

Jenkins triggers build after every change in the github repo.

The screenshot shows the Jenkins web interface for a job named 'trend-store-app-pipeline'. The job status is 'Success' (indicated by a green checkmark). The description is 'Job to build, push and deploy react app using container in k8s on AWS.' The left sidebar contains navigation links: Status, Changes, Build Now, Delete Pipeline, Configure, Rename, GitHub, Stages, Pipeline Syntax, and GitHub Hook Log. The main area displays 'Permalinks' for various build events. Below this, a 'Builds' section shows a list of builds with a filter input. The build history shows three builds: #3 (6:41 PM, Success), #2 (4:07 PM, Success), and #1 (4:01 PM, Failure).

The screenshot shows the GitHub repository settings for 'Trend-store-app', specifically the 'Webhooks' section. The left sidebar contains navigation links: General, Access, Collaborators, Moderation, and Code, planning, and automation. The main area displays the 'Webhooks' section with a description: 'Webhooks allow external services to be notified when certain events happen. When the specified events happen, we'll send a POST request to each of the URLs you provide. Learn more in our [Webhooks Guide](#).' A single webhook is listed with the URL 'http://3.235.157.253:8080/github-w...' and the event type '(push)'. The status is 'Last delivery was successful.' There are 'Edit' and 'Delete' buttons for the webhook.

Dockerhub repo: kousickd/trend-store-app

hub

ExploreMy Hub

Search Docker Hub

CtrlK

kousickd

Your personal account

Repositories

Hardened Images

Collaborations

Settings

Default privacy

Notifications

Billing

Usage

Pulls

Storage

Upgrade subscription

Repositories / trend-store-app / General

kousickd/trend-store-app

Last pushed 6 minutes ago · Repository size: 36 MB · ☆0 · ↓54

Add a description

Add a category

General

Tags

Image Management

Collaborators

Webhooks

Settings

Tags

This repository contains 3 tag(s).

Tag	OS	Type	Pulled	Pushed
3		Image	less than 1 day	6 minutes
2		Image	less than 1 day	about 3 hours
latest		Image	less than 1 day	about 3 hours

DOCKER SCOUT INACTIVE

Activate

buildcloud

Build with Docker Build Cloud

Accelerate image build times with access to cloud shared cache.

Docker Build Cloud executes builds on optimally- infrastructure with dedicated per-organization iso

Get faster builds through shared caching across y platform support, and encrypted data transfer - al

Docker commands

To push a new tag to this reposi

docker push kousickd/tre

Search

[Alt+S]

jenkins@ip-10-0-1-39:~\$ kubectl get nodes

NAME	STATUS	ROLES	AGE	VERSION
ip-192-168-21-153.ec2.internal	Ready	<none>	6h8m	v1.34.10-eks-cb19647
ip-192-168-40-63.ec2.internal	Ready	<none>	6h8m	v1.34.10-eks-cb19647

jenkins@ip-10-0-1-39:~\$ kubectl get all

NAME	STATUS	RESTARTS	AGE
pod/trend-app-deployment-5cc5f46557-b924r	1/1	Running	0
pod/trend-app-deployment-5cc5f46557-xj9kk	1/1	Running	0

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
service/kubernetes	ClusterIP	10.100.0.1	<none>	443/TCP	6h13m
service/trend-app-service	LoadBalancer	10.100.226.59	a37aee423a5314272890f7631ebf00ed-1708437506.us-east-1.elb.amazonaws.com	3000:32212/TCP	163m

NAME	STATUS	UP-TO-DATE	AVAILABLE	AGE
Deployment.apps/trend-app-deployment	READY	2/2	2	9m54s

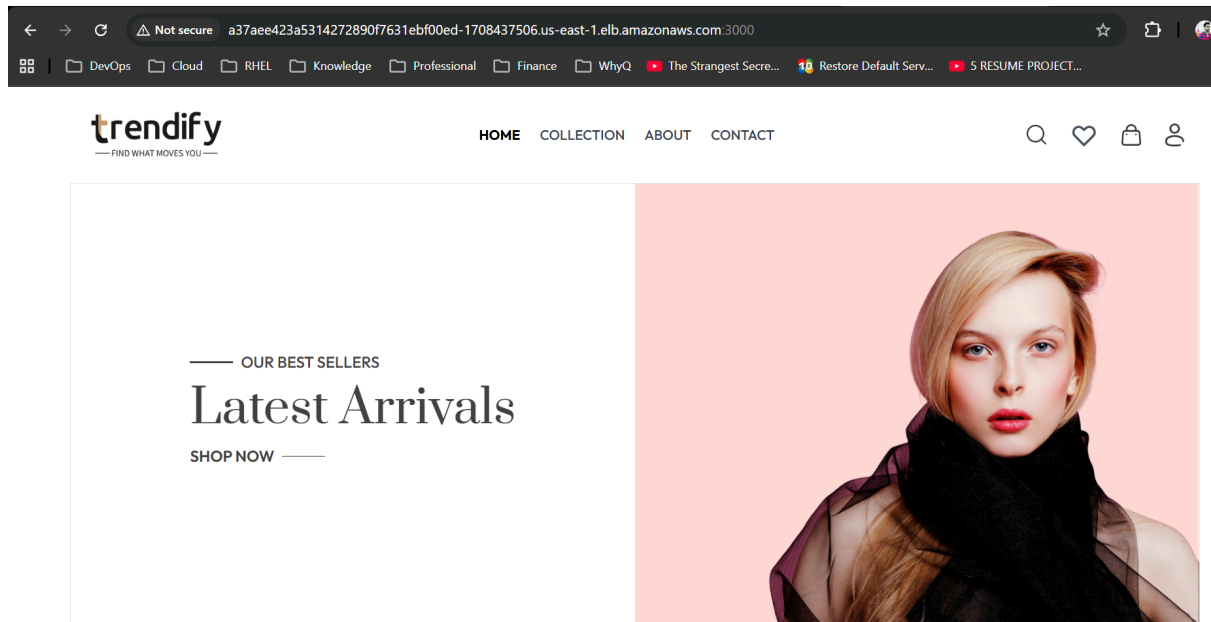
NAME	DESIRED	CURRENT	READY	AGE
replicaset.apps/trend-app-deployment-5cc5f46557	2	2	2	9m54s

jenkins@ip-10-0-1-39:~\$

jenkins@ip-10-0-1-39:~\$

Access the application in the browser via

<http://a37aee423a5314272890f7631ebf00ed-1708437506.us-east-1.elb.amazonaws.com:3000/>



5. Monitoring Setup (Prometheus & Grafana)

Deployed directly on the management EC2 instance using Docker Compose.

Navigate to monitoring directory and launch containers

```
cd ~/monitoring
```

```
docker-compose up -d
```

- **Grafana Dashboard:** Access at <http://3.235.157.253:3001>
- **Prometheus Data Source:** <http://3.235.157.253:9090>

```
aws
[Search] [Alt+S]
United States (N. Virginia)

b7a272b9937: Pull complete
0dc8fee0e5ee: Pull complete
e80cad428b4d: Pull complete
2a388505dad8: Pull complete
49c226752edd: Pull complete
8ee344dfb17: Pull complete
Digest: sha256:5ce7540c3c00ef4ab0c9d2c995c6a5b9c421f44b4a115d97a2c7af3b1c21cbb0
Status: Downloaded newer image for prom/prometheus:latest
Pulling node-exporter (prom/node-exporter:latest)...
latest: Pulling from prom/node-exporter
f5ee56b8245c: Pull complete
c3f261857c3: Pull complete
be5f3176279: Pull complete
Digest: sha256:1b4e4438fac4dd7e001dd445d161a4a2091b0fededa84093b3a8dfeae1f1be0
Status: Downloaded newer image for prom/node-exporter:latest
Pulling grafana (grafana/grafana:latest)...
latest: Pulling from grafana/grafana
c3f38fea531: Pull complete
ccdc6a4758b3: Pull complete
b38a4602e9f: Pull complete
157434f6599a: Pull complete
b3fa5aa4a9e5: Pull complete
ad2518ae2a48: Pull complete
4f4f5700e54: Pull complete
dc3f16cc959b: Pull complete
d7788a257858: Pull complete
24ad5a5d1b0c: Pull complete
Digest: sha256:f772d443e3fa0049deb2b1b30ab43242bfcfa1537614aa8d36080232cf4283
Status: Downloaded newer image for grafana/grafana:latest
Creating node-exporter ... done
Creating grafana ... done
Creating prometheus ... done
ubuntu@ip-10-0-1-39:~/monitoring$ docker-compose ps

```

Name	Command	State	Ports
grafana	/run.sh	Up	0.0.0.0:3001->3000/tcp, :::3001->3000/tcp
node-exporter	/bin/node_exporter --path. ...	Up	0.0.0.0:9100->9100/tcp, :::9100->9100/tcp
prometheus	/bin/prometheus --config.f ...	Up	0.0.0.0:9090->9090/tcp, :::9090->9090/tcp

```
ubuntu@ip-10-0-1-39:~/monitoring$
```

i-0dca6995f673f960b (Jenkins-Server)

PublicIPs: 3.235.157.253 PrivateIPs: 10.0.1.39

